P O R T F O L I O

Anuska Singh

| Ball State University | Selected Works |

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01 DESIGN AND GRAPHICS

My urban design journey, from concepts to refined graphics and development.

M U N C I E   D O W N T O W N   R E V I T A L I Z A T I O N

PROPOSED SITE INVENTORY

Richard, G.(1964). Mulberry and Charles Street intersection [Photograph].

Built in 1926, the historic Rose Court, inspired by European arcades, now faces taller neighbors.

Historic Rose Court, East Charles Street, Muncie, Indiana; digital image, December 2021, "Street View," Google Earth Pro : accessed 5 September 2023

The arcade style continues, preserving the skyline, blending timber buildings with modern architecture.

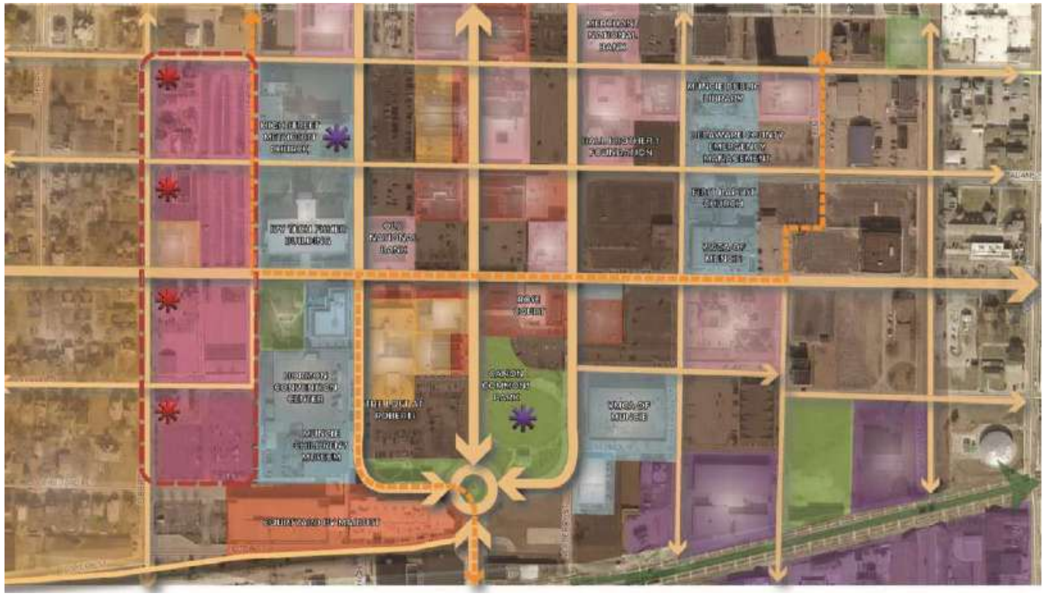
Hoosier Collections (2007). South Walnut Street from Howard Street, Muncie, Indiana [Photograph].

Walnut Street showcases Victorian, Beaux Arts, Italianate, and Colonial Revival styles with exposed wires, rail tracks, and large sign poles.

425 S Walnut Street, Muncie, Indiana; digital image, July 2023, "Street View," Google Earth Pro : accessed 5 September 2023

Many historic buildings in downtown Muncie were modernized or replaced, leaving few as heritage. Pedestrian-friendly streets lack public activities.

The site includes four downtown Muncie parking lots, aiming to become a vibrant, connected community.



Low Density 2-5 D.U./Acre (LD)
Medium Density 5-15 D.U./Acre (MD)
Commercial Office (CO)
Commercial Retail and Services (CRS)
Parks/Open Spaces (POS)
Special Use (SU)
Light Industrial (LI)
Heavy Industrial (HI)

MUNCIE DOWNTOWN REVITALIZATION

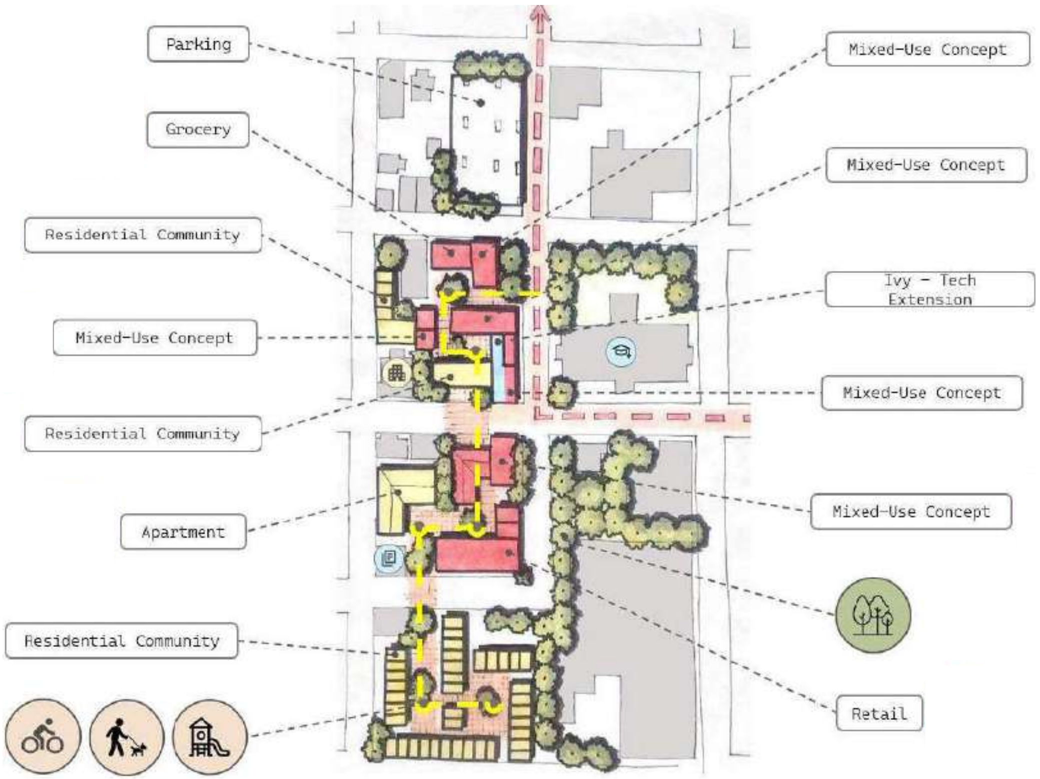
RESILIENT with essential resources—parking, eateries, and small-scale groceries.

UNITY through multigenerational interaction.

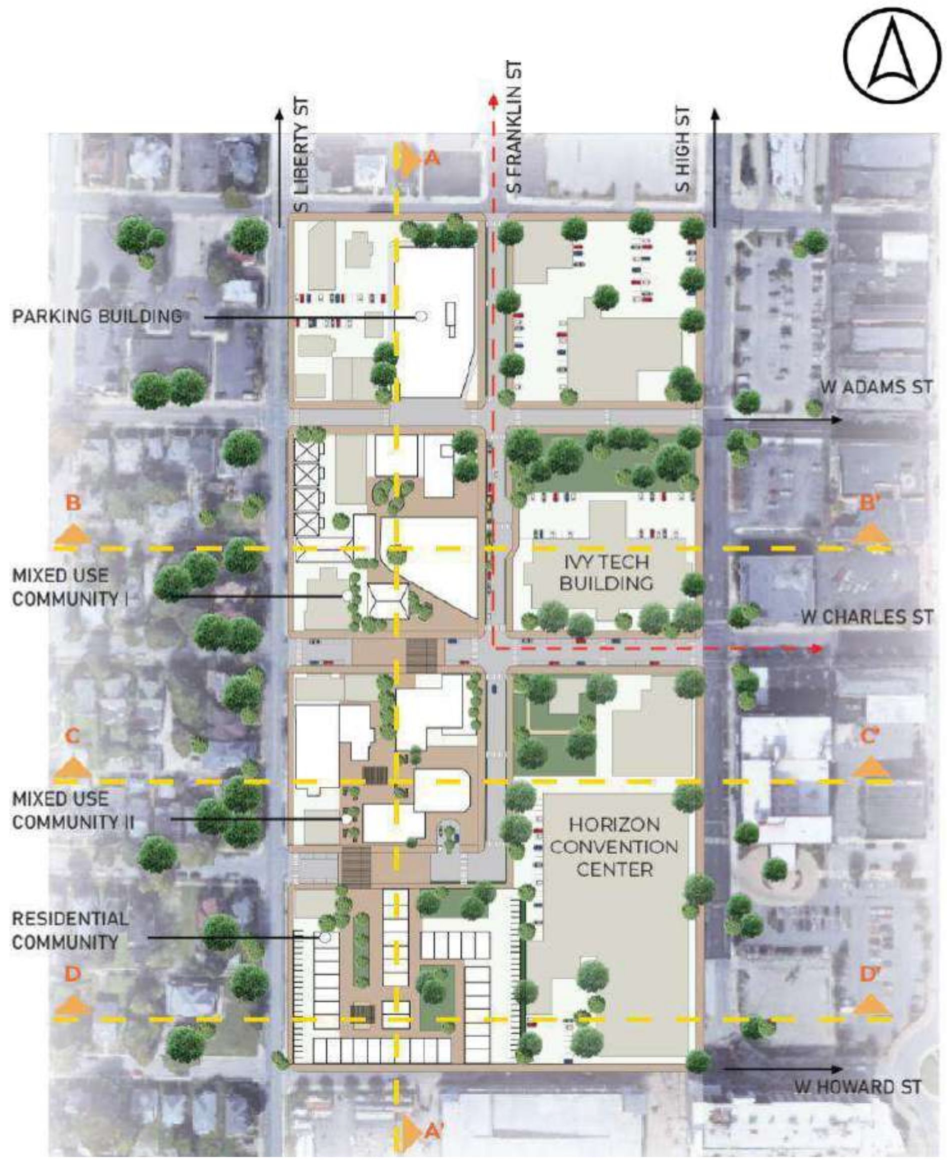
HEALTHY with safe homes and active lifestyles.

CONNECTION via housing, parks, shopping, and Ivy Tech Extension.

UPLIFTING by supporting local markets and universal access.



PROPOSED SITE PLAN

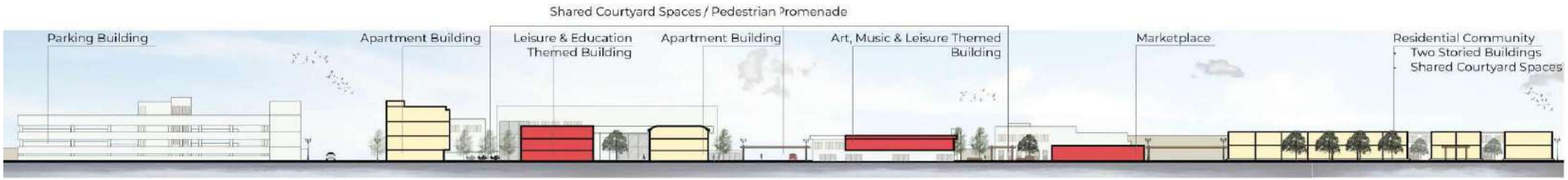


M U N C I E   D O W N T O W N   R E V I T A L I Z A T I O N

AERIAL VIEW

PERSPECTIVE VIEWS

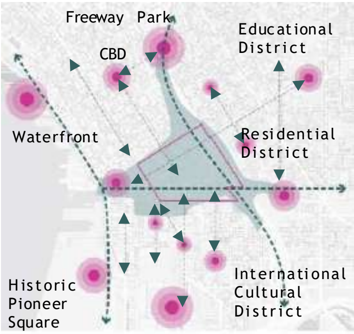
B U D
Blossoming Unity in Diversity



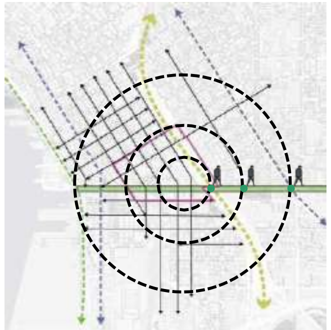
SECTION AT AA'

THE MOSAIC

SITE CONTEXT ANALYSIS



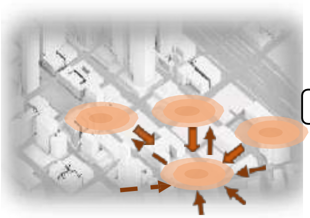
Surrounding Context



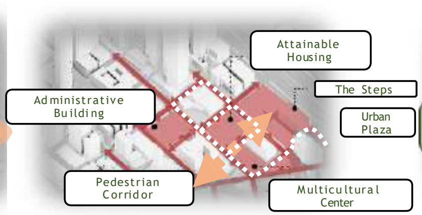
Pedestrian Activity

Green Spaces

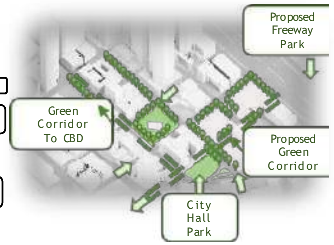
DESIGN CONCEPT



Proposed Activity Node

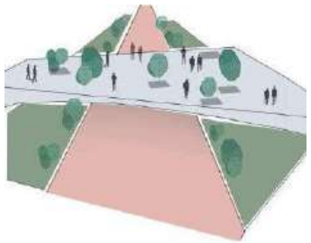


Proposed Pedestrian Connectivity

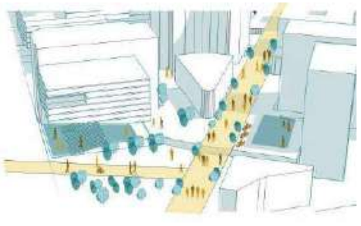


Proposed Green Spaces

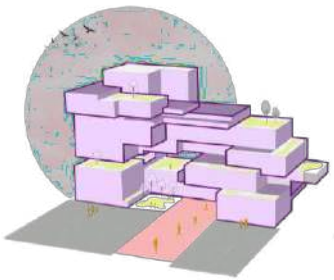
DESIGN GOALS



Enhanced User Experience



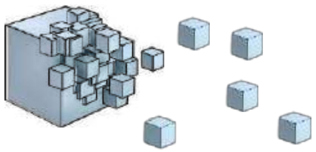
Primary activity hub

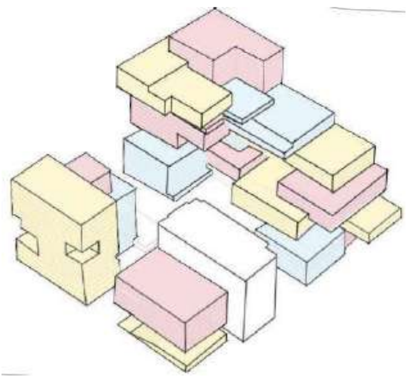


Enhancing Verdant Connectivity

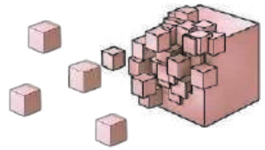
SITE PLAN

Diverse Culture

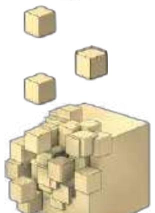




Diverse Economy



It promotes economic growth, visual connectivity, and housing through mixed-use development and sustainability.



Diverse Users

Mosaics connects people, businesses, and nature, fostering an inclusive and vibrant urban hub.

THE MOSAIC

Fall Experience at the Jefferson Pathway

Winter Experience at the Jefferson Pathway

MOSAIC transforms the area into a transit and cultural hub, driving social change in Seattle.

Its modular, replicable design revitalizes disinvested areas and supports the future Lid I-5 project.

THE MOSAIC

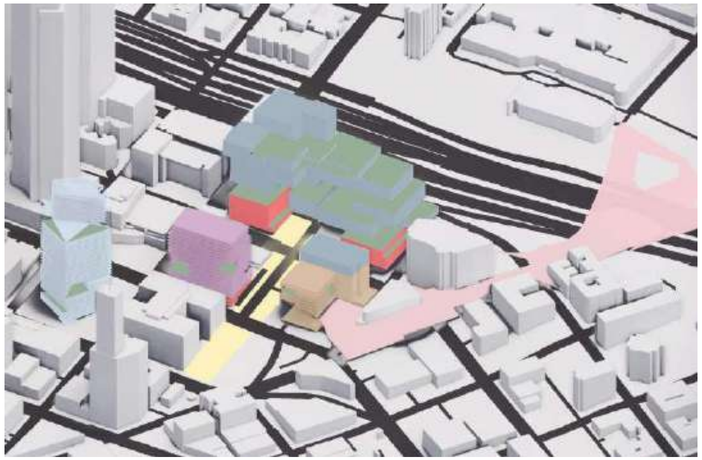
Logo Development

The logo draws from De Stijl's geometric abstraction and primary colors, while Mosaics uses small colored pieces to form intricate shapes.

AERIAL VIEW

Ideal for mixed-use development, the site boosts housing, culture, walkability, and sustainability.

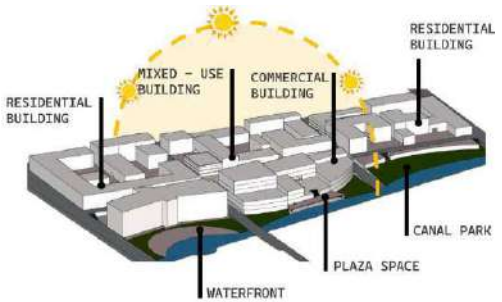
LAND USE & PROGRAM



The Mosaic: A $1.25B, 10-year, 2.5M sq. ft. mixed-use hub with housing and a green plaza.

CANALSCAPE

Canalscape transforms an area along the Indianapolis Canal Walk into a vibrant, mixed-use neighborhood, enhancing connectivity, public engagement, and waterfront access.



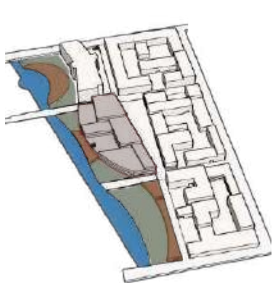
Active Nodes
Connected Spaces - Courtyards

STRENGTH
Attracts residents and visitors with a vibrant yet peaceful setting.

OPPORTUNITY
Enhances connectivity for a walkable, bike-friendly area.

WEAKNESS
Challenges in unifying underutilized spaces.

THREAT
Canal and dense structures limit expansion and integration.



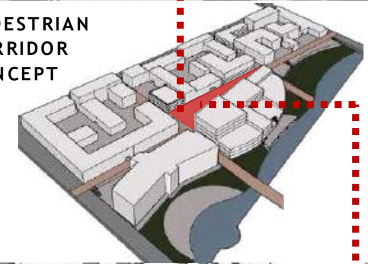
Confluence

Continuity

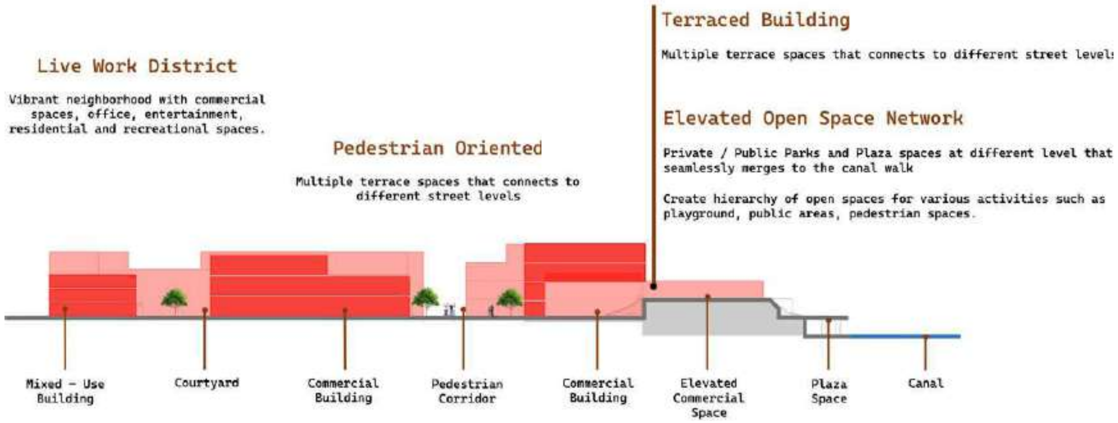
Convergence

CANALSCAPE

PEDESTRIAN
CORRIDOR
CONCEPT



SITE SECTION



$580 million
Construction cost

1,000
Parking Spaces

750
Residential Units

105,000
Office Spaces

195,000
Commercial Spaces

$49.7 million
Revenue per year

WATERFRONT

PEDESTRIAN PRIORITY

CANAL PARK

WALKING AND ACCESSIBILITY

- Pedestrian Priority
- Minimize Car Use
- Public Transport
- Overcoming Barriers

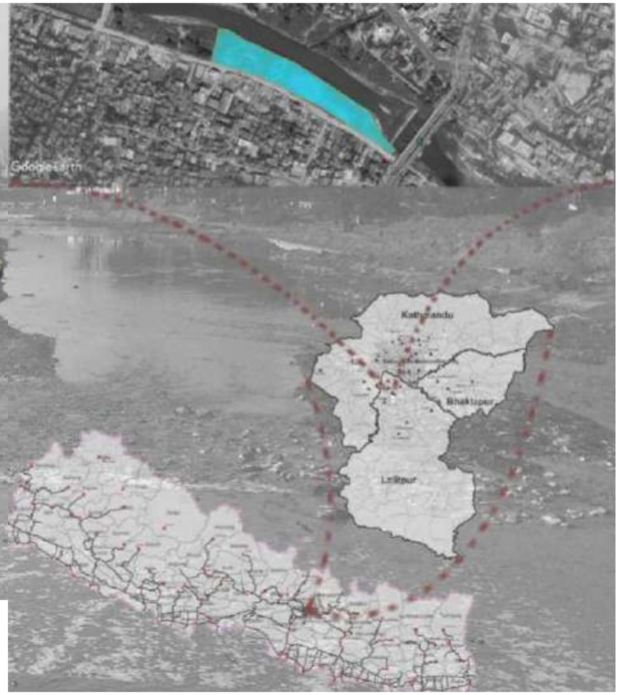
GREEN AND PUBLIC SPACES

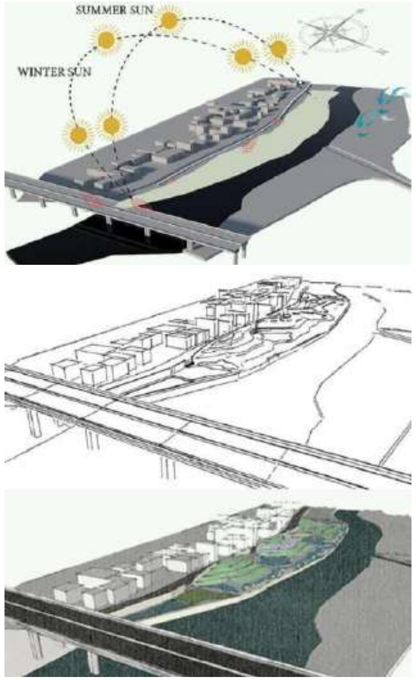
- Parks
- Plaza
- Recreation
- Eco - Friendly

CANAL SPACES

- Livable Waterfront
- Water Access
- Amphitheatre

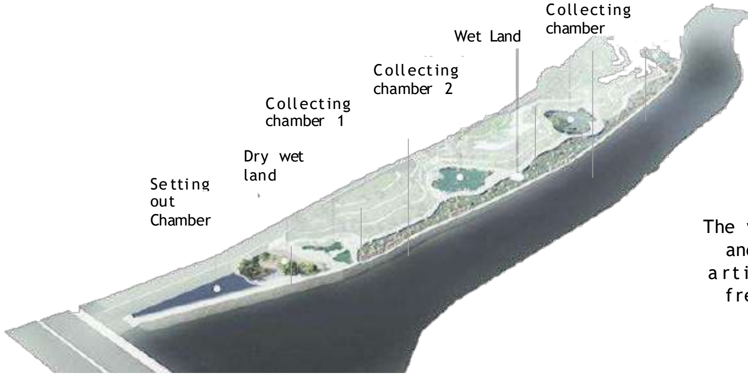
B I O P H I L I C R I V E R F R O N T





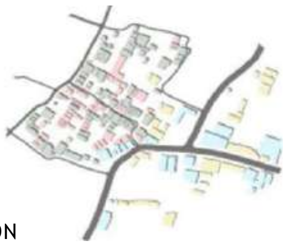
RENEW, REVIVE, REJOICE – Healing space for neighbor and nature

The design concept is derived from the ancient Chinese philosophy of Taoism that suggests to go with the flow and the natural order of things. It revolves around the idea of the Harmony of the Universe.



The water of river is purified and used for gardening and artificial pond on site for a fresh natural environment.

HOUSING AT SANKHU



BULDING DESIGNATION

COMMERCIAL ROUTE

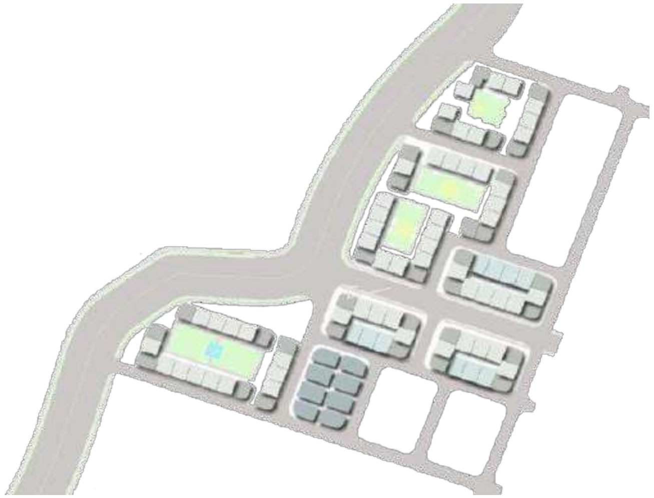
MAJOR OBJECTIVE

Preserve the heritage town's essence through planning that enhances accessibility, usage, culture, and tradition.

THE PLAZA

FOOD COURT

TYPOLOGY D



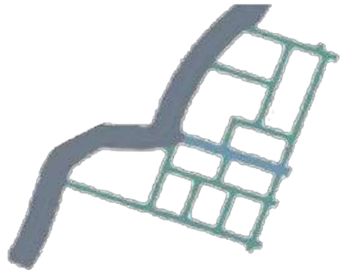
COURTYARD – WATER BODIES

The shorter side faces the street for maximum access, with three courtyards aligned to the 30m road.

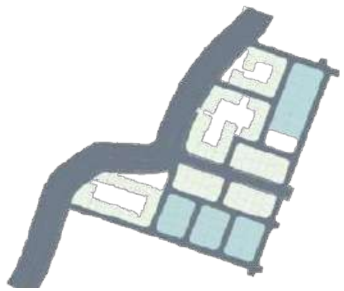
EARTHQUAKE RECONSTRUCTION

A mirror image of the roadways of the existing heritage fabric was created and placed on the other side of the road. Later, it was modified and blended into straight lines. Continuation of the existing secondary road to the expansion area - carried out.

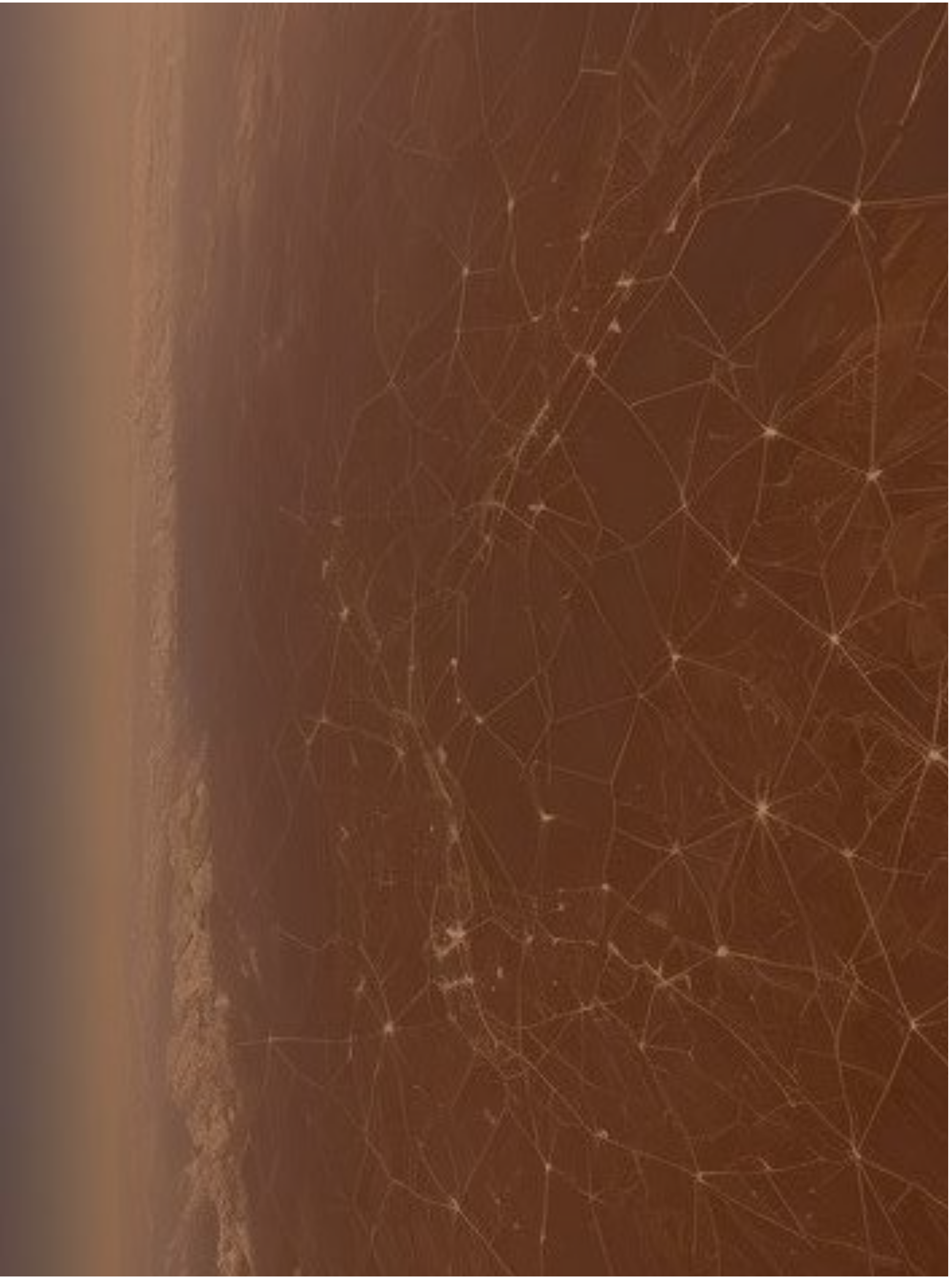
ROAD DEVELOPMENT



DIVISION OF SPACES



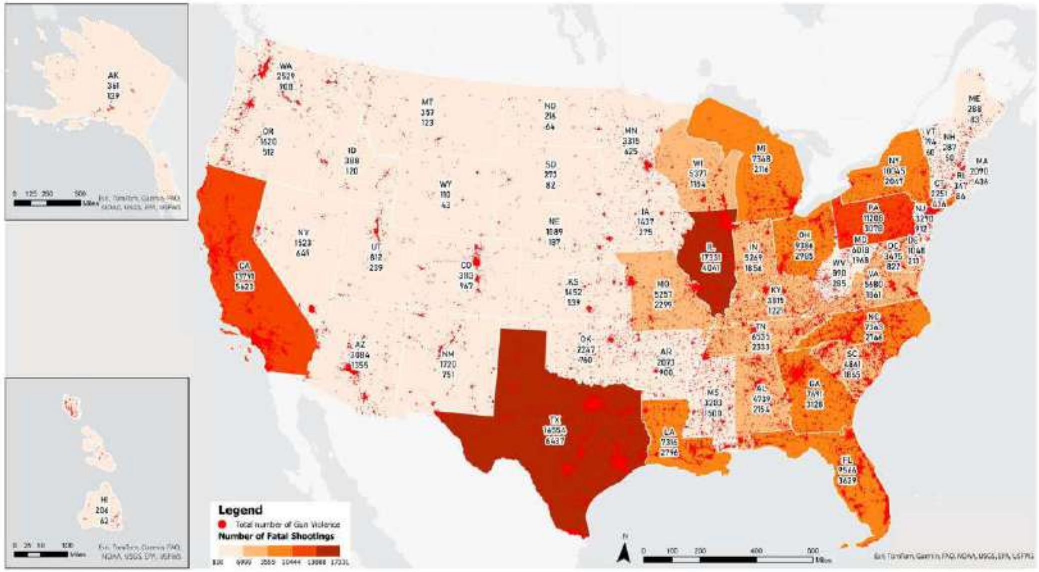
DIVISION OF PLOTS



02 GIS WORKS

GIS work featuring spatial analysis, mapping, cartography, and network modeling.

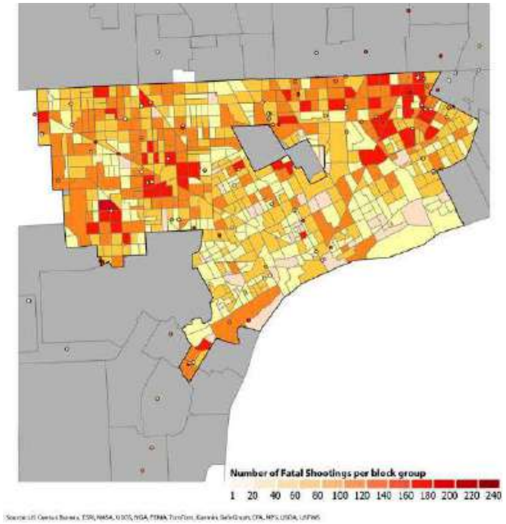
Total Gun Violence in United States



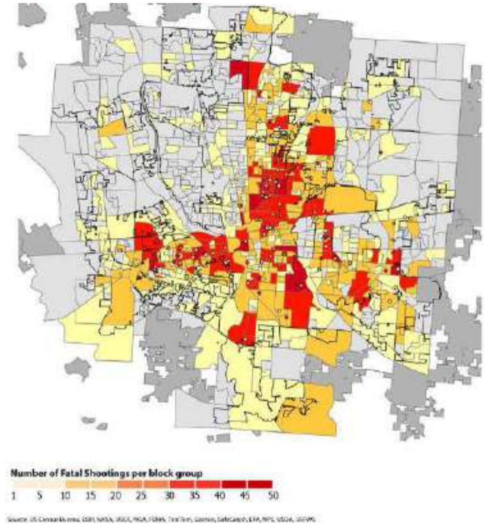
Process

- I downloaded the gun violence dataset and the shapefile for state and city boundaries from the US Census Bureau.
- Using ArcGIS Pro, I geocoded the latitude and longitude of each gun violence incident to create a shapefile. This allowed me to plot each incident on the map.
- I applied the NAD 1927 Contiguous Albers projection for the U.S. map and the State Plane projections for the four cities.
- I joined the gun violence data with the Census Block Group shapefile to aggregate the total number of incidents per block group.
- I used symbology in ArcGIS Pro to represent the density of incidents with a color gradient. For the U.S. map, I labeled states with the total number of incidents and victims, while for the city maps, I highlighted the areas with the highest concentration of incidents.
- I inserted Alaska and Hawaii in separate map frames for the U.S. map. For the city maps, I ensured a clear layout with legends and labels.

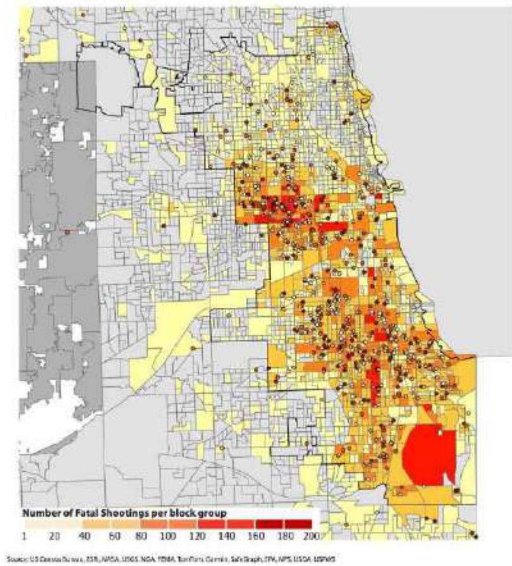
Gun Violence in Detroit, MI



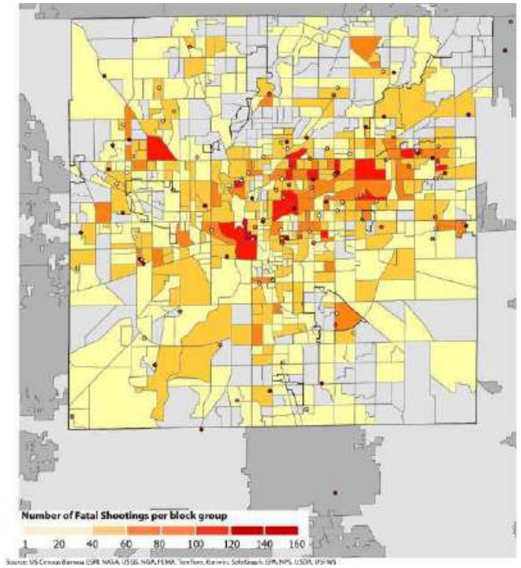
Gun Violence in Columbus, OH



Gun Violence in Chicago, IL



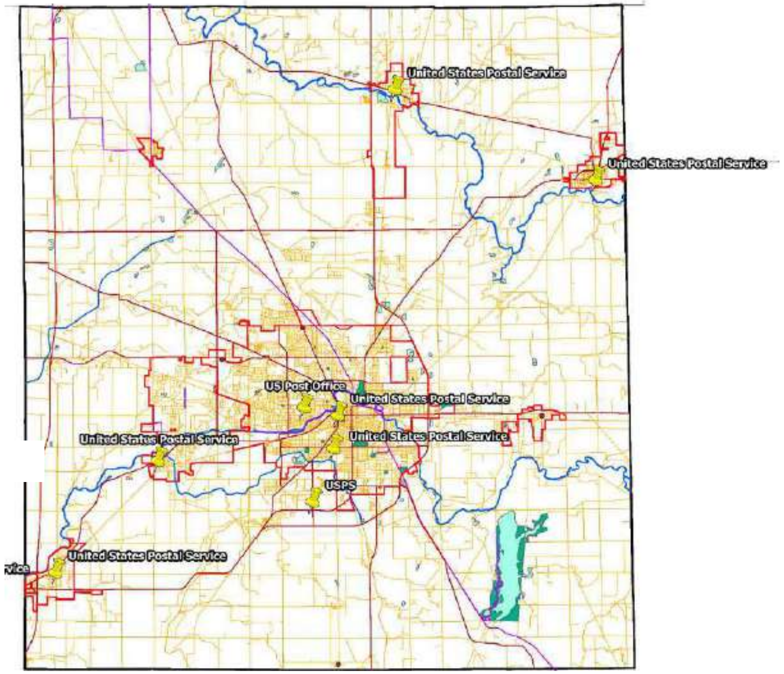
Gun Violence in Indianapolis, IN



Post Offices in Delaware County

Process

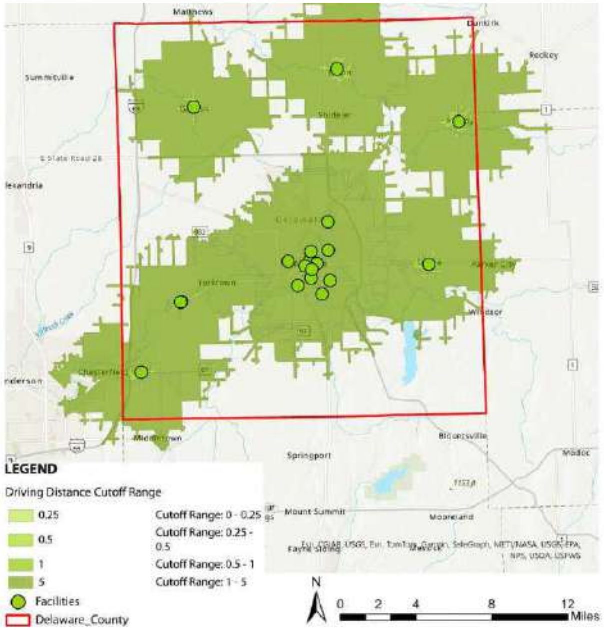
I used the US Census TIGER/Line Shapefiles for Delaware County as reference data in ArcGIS Pro and set up the address locator. Then, with the Geocode Addresses tool, I transformed location descriptions into geographic features to map the post offices.



Legend

- Municipality Boundaries
- Major Roads
- Recreation Trail
- Lakes, ponds and reservoirs
- Parks
- River, streams and creeks
- Delaware County Boundary
- Streets
- Indiana State Boundaries

Driving Distance-Based Service Area Map of Delaware County, IN



Process

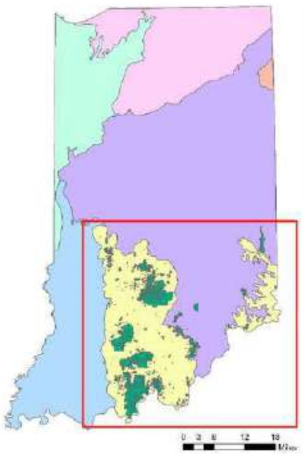
I downloaded the police station data from IndianaMap and used the shapefile. I then selected Marion County and clipped the police station data to focus on that area.

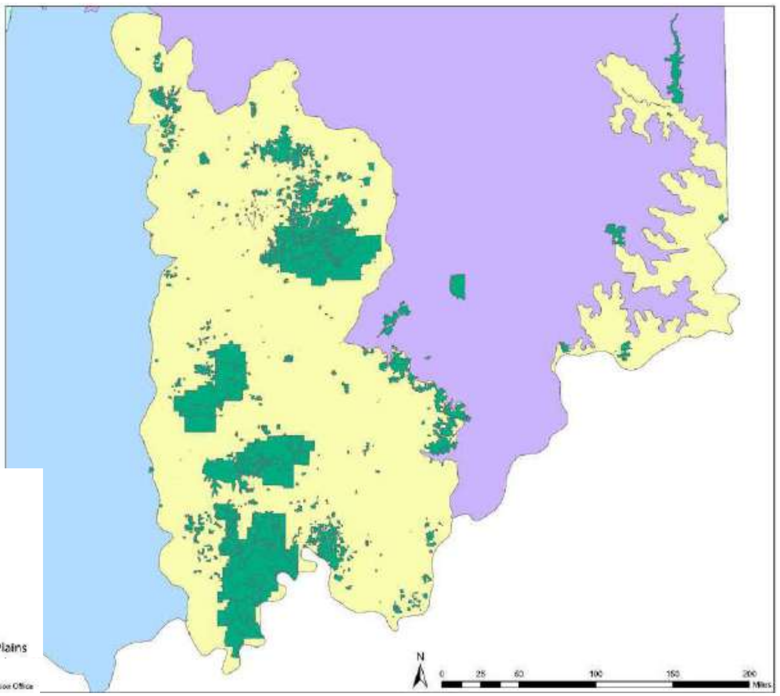
I accessed the Service Area tool from the Network Analyst extension in ArcGIS Pro and imported the police station data as input facilities.

I generated service area polygons for 5, 10, 20, and 30-minute driving times around the police stations.

I adjusted the travel settings to miles and created service areas for ¼, ½, 1, and 5-mile distances.

Parks and Recreation within Interior Plateau of Indiana





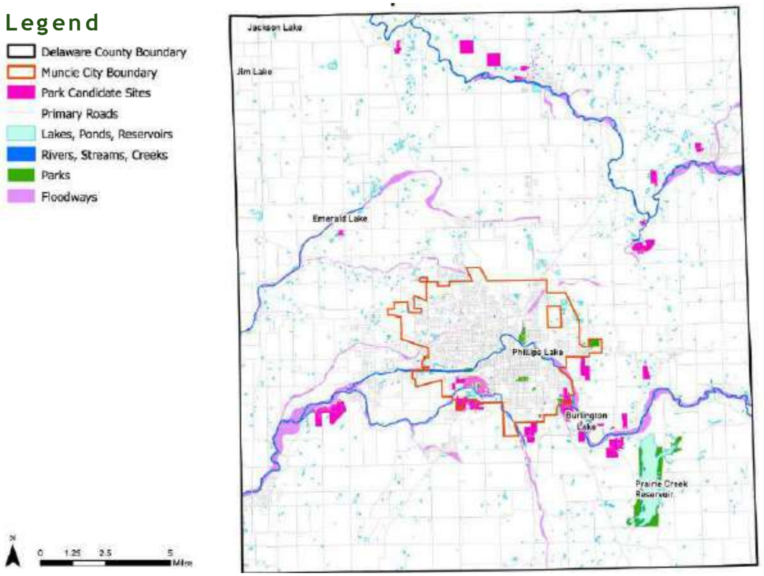
Legend

- Parks and Recreation
- Central Corn Belt Plains
- Eastern Corn Belt Plains
- Huron/Erie Lake Plains
- Interior Plateau
- Interior River Valleys and Hills
- Southern Michigan/Northern Indiana Drift Plains

Park Candidate Sites, Delaware County

Legend

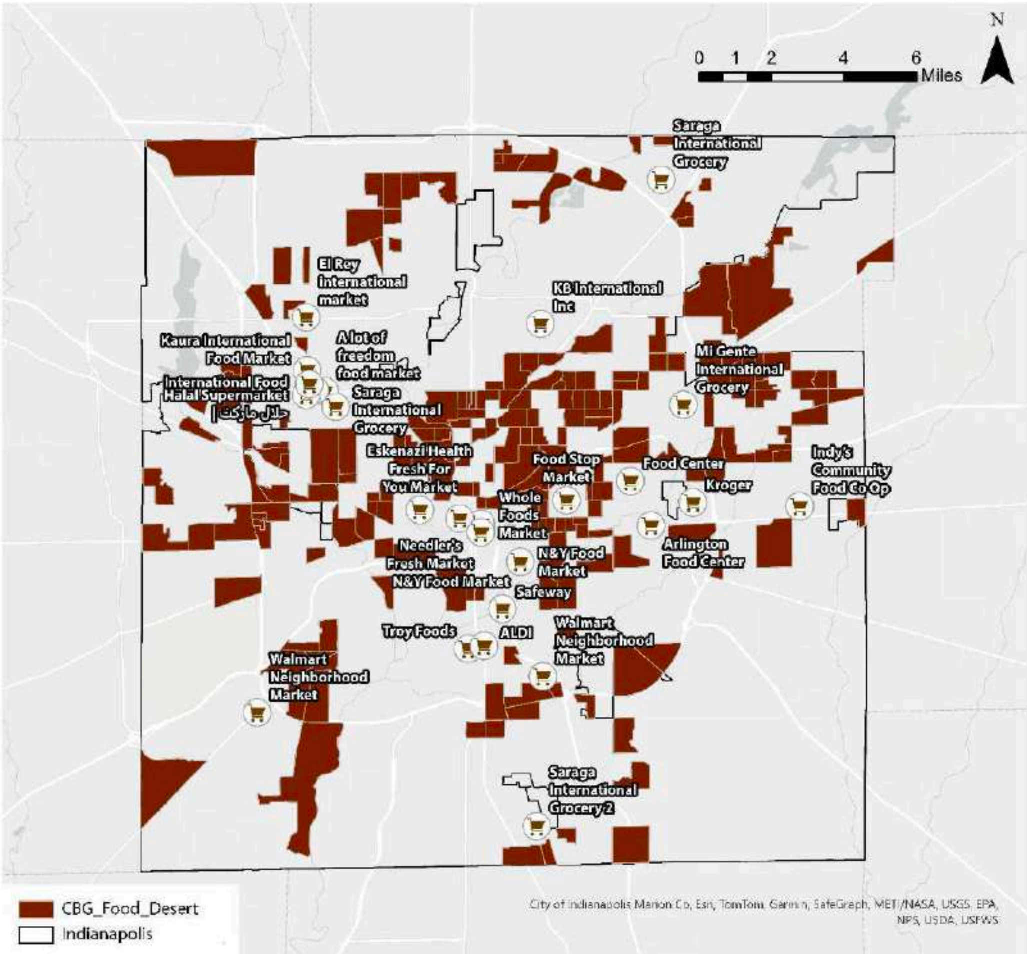
- Delaware County Boundary
- Muncie City Boundary
- Park Candidate Sites
- Primary Roads
- Lakes, Ponds, Reservoirs
- Rivers, Streams, Creeks
- Parks
- Floodways



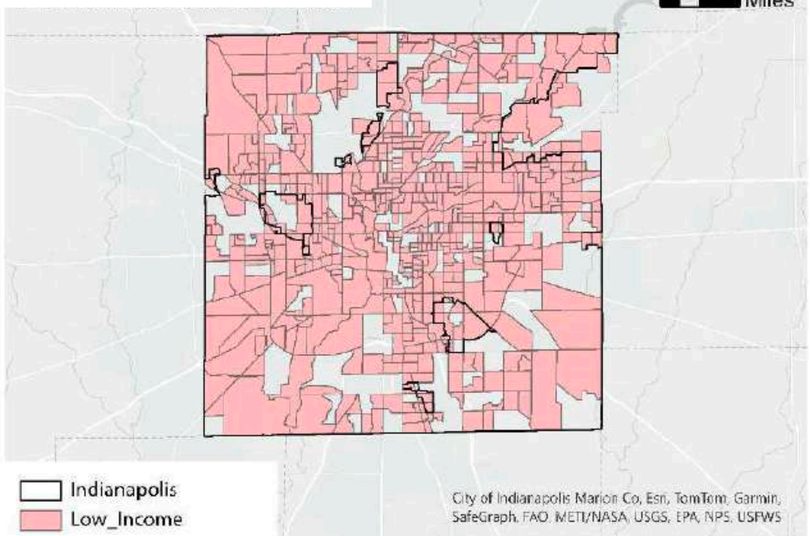
Process

I obtained data from TIGER/Line Shapefiles, IndianaMap, and the Delaware County Geodatabase. After clipping water and floodplain data to the county boundary, I selected parcels in residential or recreation zones. I created buffers for water bodies, excluded parcels in floodplains and parks, and filtered those between 10 to 15 acres.

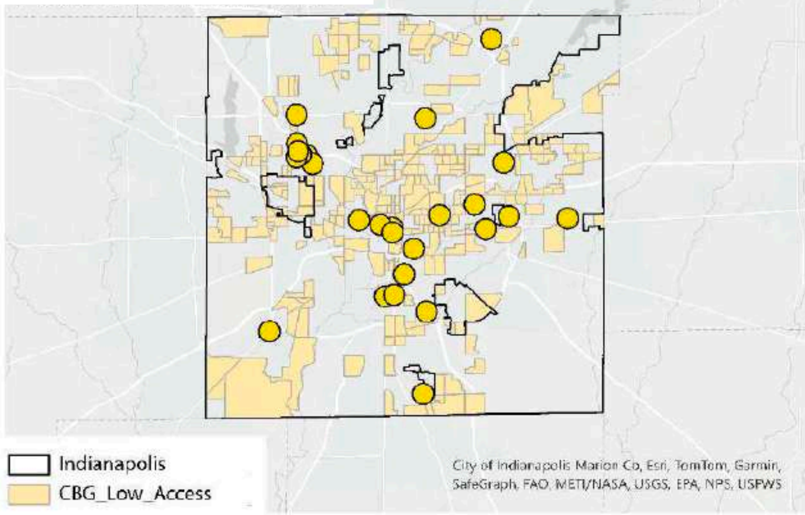
Food Desert in Indianapolis



Low Income CBG



Low Access CBG

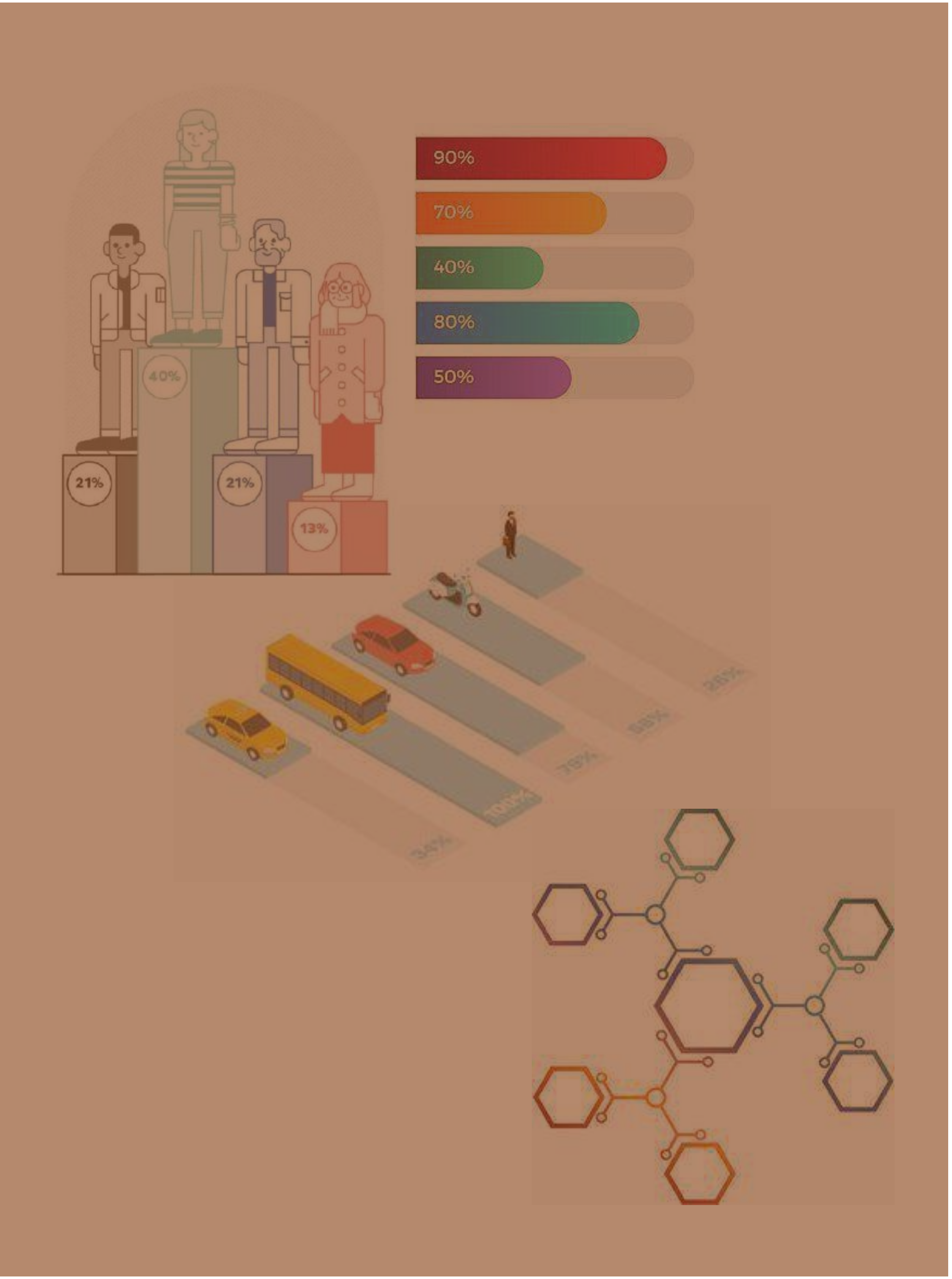


Process

I gathered Census data and USDA Food Access Research Atlas data to identify low-access Census Block Groups (CBGs) and analyze population demographics.

I geocoded grocery store locations, applied buffer analysis to measure accessibility, and used network analysis to assess travel distances along road networks.

I classified low-income and low-access areas using USDA's vehicle-based definition and performed spatial joins to integrate datasets. I visualized disparities with choropleth maps and validated results by comparing them with food insecurity reports.



03 ANALYTICAL WORKS

Urban analysis, from site studies to economic development and building proforma.

M U N C I E   D O W N T O W N   R E V I T A L I Z A T I O N

Mixed – Use Concept

Building A: 2,954 sq. ft. grocery store, $29,542 annual revenue. Building B: 6,872 sq. ft. farmer's market/retail, $68,720 annual revenue. Residential: 24 one-bed units ($158,400/year) and 8 three-bed units ($115,200/year). Parking: 39 commercial and 32 residential spaces.

Apartment

A 3-story building with 26,945.64 sq. ft. of leasable space. It includes 10 two-bedroom units (900 sq. ft. @ $750/month) and 12 three-bedroom units (1,600 sq. ft. @ $1,200/month), generating $262,800 annually. Parking: 34 spaces (1/DU + 2/DU).

Residential Community

The two-story building has 50,645 sq. ft. of leasable space, with 46 one-bedroom units ($550/month) and 28 two-bedroom units ($750/month), generating $555,600 annually. It also offers 74 parking spaces.

Parking

300 parking = 3 floors

Mixed – Use Concept

The 1-2 story retail building has 7,452 sq. ft. generating $74,520 annually, with 29 parking spaces. The Ivy Tech Extension on the second floor generates $26,176 annually from 2,617.67 sq. ft. The residential section has 10 two-bedroom units, generating $90,000 annually, with 50 parking spaces.

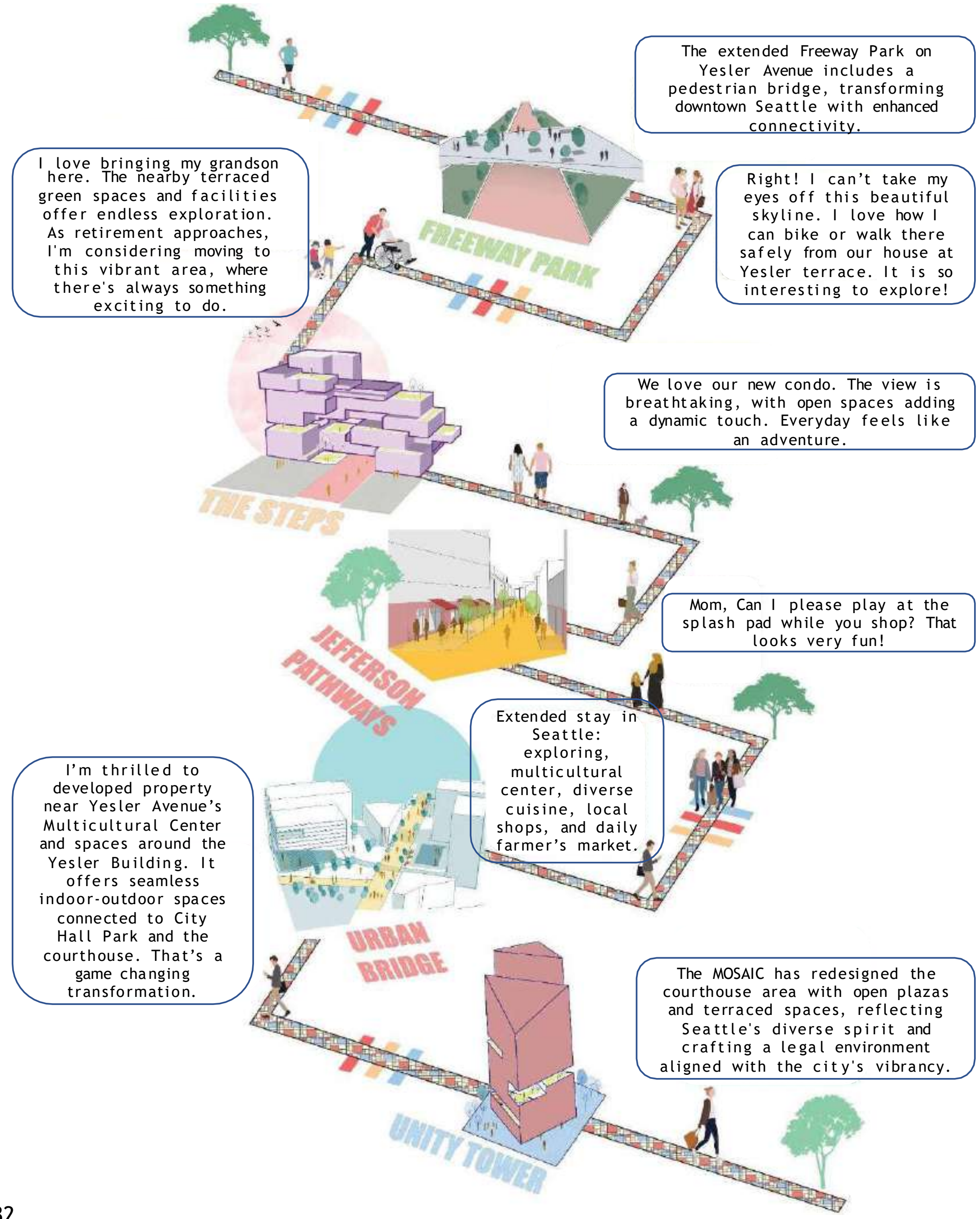
Mixed – Use Concept

Building A (Restaurant) has 7,461.66 sq. ft. of leasable space, generating $74,616 annually at $10 per sq. ft. Building B (Retail) has 5,151.62 sq. ft., generating $51,516 annually. The property includes 84 parking spaces (1 per 100 sq. ft.).

Mixed – Use Concept

The two-story retail building has 21,044 sq. ft. of leasable space, generating $210,440 annually at $10 per sq. ft., with 84 parking spaces (1 per 250 sq. ft.).

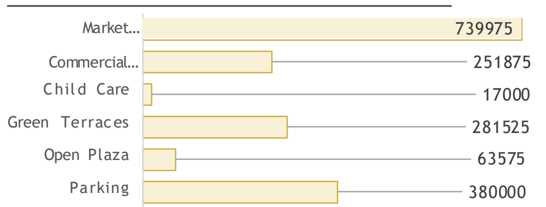
T H E   M O S A I C



TOTAL UNLEVERED IRR: 25.46%, TOTAL LEVERED IRR: 42.5%

PHASE - 1

Program



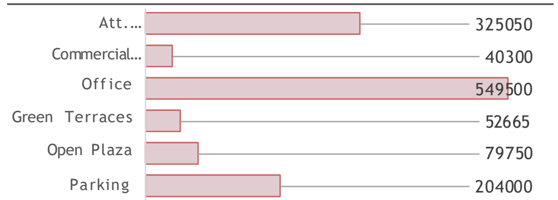
UNLEVERED IRR: 42%

LEVERED IRR: 61%

Total Buildout: 1,604,375.00 SF
Total Development Cost: $ 600,841,332

PHASE - 2

Program



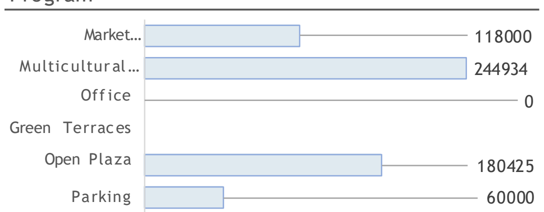
UNLEVERED IRR: 5%

LEVERED IRR: 11%

Total Buildout: 1,251,265.00 SF Total
Development Cost: $476,307,344

PHASE - 3

Program



UNLEVERED IRR: 49%

LEVERED IRR: 23%

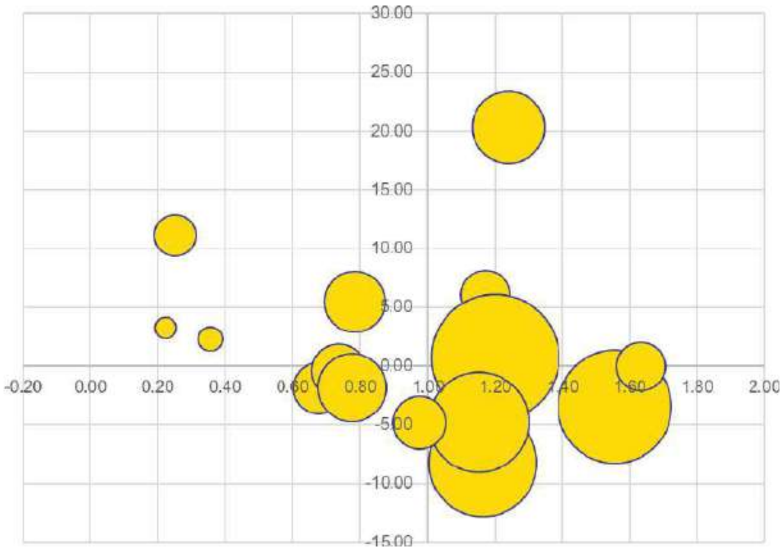
Total Buildout: 1,098,495.00 SF
Total Development Cost: $ 169,220,422

ECONOMIC DEVELOPMENT

Madison County, in central Indiana's Anderson MSA, saw population decline. Anderson grew with railroads and natural gas, later shifting to auto manufacturing.

City of Anderson Cluster Analysis (2018-2023)

Location Quotient



Economic Profile

Anderson's population declined 22.6% since 1970, with high housing vacancy (17.5%) and crime rates. Education and income lag behind state averages, with a 23% poverty rate. Despite economic decline, the city offers cultural amenities and strategic transportation infrastructure.

STRENGTH

Anderson boasts excellent amenities, infrastructure, and quality employers, supporting quality of life and economic growth.

OPPORTUNITY

Regional connections & available land offer potential for growth and development in various industries.

WEAKNESS

Anderson faces historical disinvestment, declining population, and educational challenges.

THREAT

Economic uncertainties, post-industrial decline, & competition for skilled workforce.

Vision Statement

Anderson will foster economic growth by honoring its industrial legacy while embracing technology, sustainability, and resident well-being. Through job diversification, innovation, and quality-of-life investments, the city will thrive, attract new talent, and retain local assets.

Goals and Strategies

Goal #1: Diversify and expand employment opportunities

Goal #2: Develop talent and connect residents to satisfying careers

Goal #3: Spur innovation and entrepreneurship

Goal #4: Foster a vibrant and inclusive community through investments in quality of life

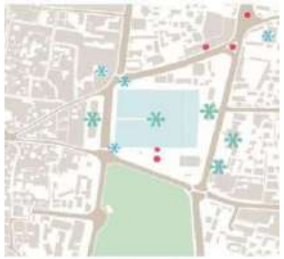
Goal #5: Improve quality of place by encouraging housing development and community revitalization.

R A N I P O K H A R I

FIGURE AND GROUND

ROAD NETWORKS

LANDMARKS



PEDESTRIAN MOVEMENT

FIGURE AND GROUND

Centrally located open space with compact settlements on the periphery and unmanaged, crowded areas. Black denotes buildings, white denotes open spaces.

GREEN SPACE ANALYSIS

Unmanaged tree placement with minimal space for long-term growth.

ROAD/ ALLEY NETWORK

Depicts the street hierarchy within the built environment, with small alleys at building frontages featuring improper streetscapes.

NODES & JUNCTIONS

Higher activity intensity occurred at four major nodes, causing traffic congestion from converging paths.

LANDMARKS

Major Landmarks: Rani Pokhari, Saraswati sadan, Durbar High School, Ghantaghar, Mosque ; Minor Landmarks: Nachghar, Ganesh Mandir, Seto Dhoka

PEDESTRIAN MOVEMENT

Red spots denote stationary pedestrians, while blue spots denote moving pedestrians.

ACTIVITY CHART

High activity intensity at nodes and areas with shops and bus parks.

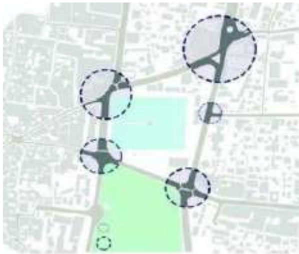
BUILDING TYPE : STOREY

Buildings with various functions, height, façade were found. Number of storey was also one of the major determining the aesthetics of the area.

8 Storeys
7 Storeys
6 Storeys
5 Storeys
4 Storeys
3 Storeys
2 Storeys
1 Storey

GREEN SPACE ANALYSIS

NODES AND JUNCTIONS



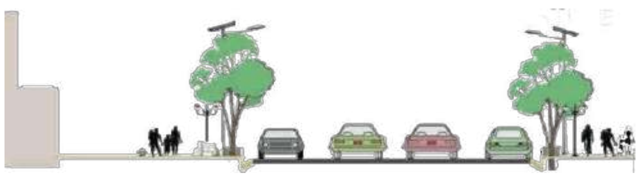
ACTIVITY CHART

BUILDING TYPE: STOREY

MASTERPLAN: RANIPOKHARI

DETAIL PLAN AT A

STREET SECTION



1. PUBLIC SPACE – PARK

2. AMPHITEATRE

4. URBAN FURNITURES

6. VIEW FROM PAVILLION

04 MISCELLANEOUS

W R I T I N G

Rehabilitating squatter settlements

Begin with affordable housing solutions

The Kathmandu Post, January 05, 2023

On December 4, Wednesday, the Kathmandu Metropolitan City Office implemented an eviction plan for the Bagmati Corridor squatter settlements. The action met with resistance from the residents of these communities and sparked a debate among intellectuals about the morality of the actions taken by both sides. While there is consensus that the long-standing squatter settlements need to be correctly rehabilitated and removed from the Bagmati Corridor, the main question remains: how can this be done fairly and equitably?

The urban population has increased significantly compared to the rural areas. Projections indicate that the population will continue to grow until around 2050. The inability of governments to keep up with the rapid pace of urbanization and provide affordable housing has increased slums and squatter settlements.
When asked how squatters see themselves, many representatives of squatter settlements stated that they do not “capture the land”, but they save the land”. They described that if the squatters had not been there, the private sector would have bought the public land.

The general public often perceives squatters negatively, particularly in developing countries like Nepal, where land is valuable. There is also a perception that many squatters are wealthy and own land in their villages. However, it is essential to remember that these perceptions are only sometimes accurate and do not represent all squatters. It is also necessary to recognize that the voices of strong community figures may be louder and visible in the media while silencing the voices of real squatters due to their perceived weakness.

The government may view squatter settlements as illegal because they build their shelters on land owned by the government without proper legal documentation or permits. By taking the stance that squatter settlements are illegal and not taking responsibility for their residents, the government sends a message that these communities are not valued and do not deserve support. This attitude only serves to widen the gap between the government and the landless, making it more difficult to find solution for rehabilitating these settlements.

The proximity to income-earning opportunities in the city is often a crucial issue for the urban poor. Squatters build their shelters on public land and other environmentally-sensitive areas like riverbanks or flood plains, steep slopes and vacant spaces under high-voltage electrical transmission lines. To gain access to land for housing, the urban poor usually sacrifices tenure security. Situations force them to illegally infringe on any vacant land (often ill-suited for accommodation).

Due to lack of affordable land, and housing, the migrant population squats. In Nepal, squatting can be done openly or covertly by individuals or groups. Squatters often use scavenged materials such as plastic and corrugated metal to construct shelters or reside in abandoned buildings. Several factors contribute to the rural exodus, including push factors such as caste segregation, social exclusion of widows and the Dalits, and uncertain economic conditions, as well as pull factors such as the availability of better jobs, healthcare, education and other amenities. This phenomenon has led to several problems, including the continuation of informal settlements, a widening gap between wealthy and impoverished individuals, a lack of tenure security and unsustainable land use patterns.

Even though it may seem that squatters occupy valuable land in the city center, they still face several problems. The fundamental issues inherent in slums are health risks. The settlements are generally along the river, polluted, and at risk of flooding. Most use open-sky toilets or drains connected directly to the river. The residents have immediate impacts on health and the environment, namely infant mortality, diarrhea, and respiratory diseases.

Further, the squatter environment also faces cultural problems and social exclusion. Squatter populations rely primarily on daily wages and small businesses for their income. High unemployment often leaves people frustrated. These lead to domestic violence, including alcohol or drug abuse. These settlements’ lack of shelter and privacy can lead to increased exposure to sexual violence. Regardless of its form, tenure is often registered in the name of men, leaving women dependent on their male relatives for tenure security. The quality of housing profoundly influences children’s health, educational advancement, and general well-being. It can profoundly impact children’s growth, including the right to education, health and personal security. The poverty alleviation strategy and the current housing development trend have failed to meet the housing needs of the urban poor living in slums and squatter settlements. The squatter settlements were established on the Bagmati river bank over a long time, and some have lived there for 20 years.

By disregarding the squatter settlements, the government is effectively disincentivizing the development of these communities and making it more challenging to address the potential solutions. Instead, it would be more effective to adopt a more proactive and inclusive approach that seeks to understand the root causes of informal settlements and works to find solutions that benefit both the government and the landless.

Providing housing solutions that are affordable, safe and appropriate for the needs of squatter settlement residents can be an essential step in improving living conditions and reducing the risk of increment of squatters. Overall, addressing the challenges faced by squatter settlements in Kathmandu will require a combination of approaches, including providing basic infrastructure and services and regularizing land tenure, among others.

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Selected Works 2016-2024